Statement of Purpose

Eric Schneider (<email censored>)

To whom it may concern, I intend to apply for the Doctor of Philosophy (PhD) program in Computer Science for Virginia Tech, to start in fall 2024.

Background and Qualifications

I am currently pursuing my Master's (MS) in Computer Science at UNC Chapel Hill, and am slated to graduate in May 2024. I graduated with my Bachelor of Science in Computer Science and in Mathematics at UNC Chapel Hill too in May 2023.

My coursework inspired me to delve deeper into systems, my subfield of interest, and to take advanced classes and branch out into research. Under Professor Donald Porter, I contributed to a paper, *Adaptive and Efficient Dynamic Memory Management for Hardware Enclaves*, which has been submitted to EuroSys '24. Our paper describes techniques to efficiently use EDMM, a feature of Intel SGX2, to achieve faster and more flexible enclave applications. While the original version of Intel SGX (SGX1) only supports fixed-sized enclaves, SGX2 allows enclaves to be dynamically resized. However, dynamic allocation of enclaves significantly degrades application performance; thus, we use traditional systems techniques, such as pre-allocation, batch allocation, and lazy free, to achieve performance on par with or exceeding applications using SGX1. For this paper, I mainly contributed writing, benchmarking, and some literature review.

Secondly, I have been a part of another research group, AD∆ER, under Professor Ketan Mayer-Patel, Professor Montek Singh, and Andrew Freeman. Our group has developed an alternative event-based video representation inspired by event-based cameras, with associated transcoders and tools written in Rust. While traditional video is represented as a series of frames separated at a fixed time interval, event-based video is represented as a series of interleaved events. Each event contains the location of the pixel that produced that event, the time, and a measure of intensity. Other event-based video formats exist, but they are tied to specific cameras and often proprietary; meanwhile, AD∆ER is camera-independent and is open source. AD∆ER is amenable to compression and feature detection, which we are exploring. I have contributed new features to AD∆ER, such as a transcoder bandwidth limiting feature, better cross-platform support and CI, and am in the process of writing a paper, *AD∆ER: an Event Video Representation and Tool Suite*. This paper will detail the architecture of the project, which will be submitted to the MMSys '24 "Open-Source Software and Datasets" track.

I have also interacted with research through creating NeuroRuler, a tool used by neuroscientists at UNC to calculate head circumference from MRI scans for research. I have taken a few research-focused graduate classes, and I have been working on a research project to port Keystone, an open source trusted execution environment for RISC-V, to uClinux. I have also enjoyed interacting with research through serving on the ASPLOS '24 Artifact Evaluation Committee.

Outside of research, I have served as an undergraduate/graduate teaching assistant for seven semesters, for two classes, System Fundamentals (an intro-level class mainly composed of underclassmen) and Operating Systems (an advanced elective mainly composed of upperclassmen), under four different professors. In this capacity, I have tutored students in

office hours, graded coursework, and managed quizzes, programming labs, midterms, and the final. A lab that I created from scratch, the MIPS Simulator lab, has been completed by ~1000 students over the 5 semesters it's been offered. Outside of the university setting, I was an intern at VMware for their Telco Cloud Platform for two summers, contributing to microservices that process >1 million messages a second in production. As a hobby, I have worked on a few Minecraft mods, some of which have millions of downloads, and I have also served as an administrator for the largest modded Minecraft wiki.

Goals

As a PhD student at Virginia Tech, I hope to focus on conducting research in the systems field, especially in operating systems and architecture. While I am not married to any subfield of a subfield, I find the confidential computing space to be especially interesting, and I imagine the model of mutual distrust between supervisor software and application to be a likely future of computing. More generally, I am interested in efficient use of new hardware technologies and architectures (such as Intel SGX/TDX, tiered memory, hardware accelerators, and RISC-V), as well as improving the safety, reliability, and performance of software and hardware systems.

As a burgeoning researcher, I want to collaborate with experienced researchers from both academia and industry. However, I also would like to work with aspiring researchers, like undergraduates and Master's students, and serve as an effective mentor to them. I've had some great student mentors, as well as a couple that were less effective. I've learned from both, even if in different ways. In a notoriously intimidating field like systems, I want to make junior colleagues feel welcome and empowered to pursue research and achieve their own goals, whether they plan to pursue a PhD themselves or head directly to industry.

While I want to focus on research during the duration of my program, I hope to use required coursework to improve my background. I plan to take courses in networks or distributed systems, as well as cryptography or security. Regrettably, there are close to zero graduate-level architecture classes at UNC, so I hope to take at least one at Virginia Tech. I also want to take a machine learning or AI class; although ML/AI isn't my preferred subfield, I don't think I could call myself a great computer scientist if I don't have any background in the fastest growing area of our field.

Long-term Goals

Upon completion of a PhD, I hope to conduct high-impact systems research in industry. Advancing the subfield cuts through the whole stack, and thus can have an oversized impact. For example, the recent major advancements in AI owe as much to advancements in computer architecture as they do to improved machine learning algorithms. Safer, faster, and more reliable software and hardware systems positively affects everyone who uses computing systems, from personal computers, to mobile devices, to IoT devices and nodes in a cloud computing environment. I believe the work of getting a PhD will best position me to make such an impact.